ORIGINAL ARTICLE

Efficacy and safety of interleukin-17 antagonists in patients with plaque psoriasis: a meta-analysis from phase 3 randomized controlled trials

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Abstract

Background The interleukin-17 (IL-17) cytokine pathway plays a key role in the development of psoriasis. Antibodies targeting IL-17 or blocking its receptor may be a new therapeutic approach for psoriasis. To assist treatment selection in daily practice, it is essential to understand the benefit and risk profile of IL-17 antagonists.

Objective We performed a meta-analysis to evaluate the efficacy and safety of IL-17 antagonists in patients with psoriasis.

Methods We searched a number of databases for relevant randomized controlled trials (RCTs) published before May 2016. The following outcomes were evaluated: Psoriasis Area and Severity Index (PASI) 75, 90, 100 response, Investigator's Global Assessment (IGA) score of 0 or 1 response, adverse events (AEs) and withdrawals. The meta-analysis was performed using Review Manager 5.2 software.

Results Nine RCTs with 5951 patients were included. IL-17 antagonists achieved higher PASI 75, 90, 100 response rates and Dermatology Life Quality Index 0 or 1 response rates than placebo and a lower incidence of discontinuations due to lack of efficacy. In the safety analysis, no significant differences were found between the IL-17 antagonists and placebo in the proportion of patients with serious AEs, cardiovascular disease and discontinuations due to AEs. However, IL-17 antagonists were associated with a higher proportion of patients with any AEs and infections than placebo.

Conclusion IL-17 antagonists were effective, with an acceptable safety profile, for patients with plaque psoriasis. Vigilance because of the potential for infection will be necessary for IL-17 antagonists.

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Conflicts of interest

The authors declare that they have no competing interests.

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Introduction

Plaque psoriasis is a chronic, inflammatory, immune-mediated skin disease affecting about 1–3% of the world's population.[1] Currently almost incurable, psoriasis is frequently accompanied by impairment of life quality,[2] several complications and dissatisfaction with therapy.[3] Traditional oral treatment for plaque psoriasis has been limited due to intolerable side effects and organ toxicity.[4] Through our understanding of psoriasis pathogenesis, a number of available biologics under development that target cytokines are increasingly receiving attention.[5–10]

An abnormal T-cell-mediated immune response plays a major role in the process of plaque psoriasis.[11] Type 17 helper T cells (Th17 cells) produce IL-17A, which is highly expressed in psoriatic lesions and is thought to be an important cytokine in the pathogenesis of psoriasis.[12–16] Keratinocyte differentiation is extensively altered in plaque psoriasis.[17] Besides IL-17A, various other cytokines, including IL-23, IL-21, IL-22 and TNF-α, produced by keratinocytes are present at highly increased concentrations in psoriatic skin.[18] The cytokines IL-23 and IL-1β are able to produce large amounts of IL-17A in the skin, and this leads to disease progression by driving Th17 cell differentiation from naive T cells[19] and stimulating gamma delta T cells.[20] Preclinical studies indicate antagonizing the IL-17 pathway is a

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worthwhile therapeutic target.[21] The newest and most promising biologic therapies for psoriasis are antibodies targeting IL-17 or blocking its receptor.[22,23] These primarily include anti-interleukin-17A monoclonal antibodies (anti-IL-17A mAb), secukinumab (AIN457) and ixekizumab (LY2439821), and an anti-interleukin-17 receptor A monoclonal antibody (anti-IL-17RA mAb), brodalumab (AMG827), which are collectively referred to as 'IL-17 antagonists' in the following discussion.[24]

In recent years, phase 3 randomized controlled trials (RCTs) of IL-17 antagonists for the treatment of psoriasis have been reported. To provide an up-to-date picture of the clinical efficacy and safety of IL-17 antagonists in psoriasis, we conducted a meta-analysis.

Methods

Eligibility criteria

Studies included in this meta-analysis met the follow criteria: (i) they were phase 3 RCTs that enrolled patients with plaque psoriasis; (ii) their duration of treatment was ≥12 weeks; (iii) they used a parallel design or crossover design of IL-17 antagonists vs. placebo; and (iv) their data reported information on Psoriasis Area and Severity Index (PASI) response, Dermatology Life Quality Index (DLQI) score of 0 or 1 response, adverse events (AEs) and discontinuations.

Study identification

We searched all relevant studies published before May 2016 on the electronic databases MEDLINE, Embase and Cochrane Library, using the following search terms: 'psoriasis', 'secukinumab', 'brodalumab', 'ixekizumab', 'anti-IL-17', 'IL-17 inhibitor' or 'IL-17 antagonists'. Additionally, Center Watch Clinical Trials Listing Service (http://www.centerwatch.com), Current Controlled Trials Service (http://www.controlled-trials.com) and clinical trials registered at ClinicalTrials.gov (http://clinicaltrials.gov) were searched for details of any relevant clinical trials in progress.

Study selection was performed by two independent investigators. First, titles and abstracts of cited articles were evaluated by two investigators to determine their relevance. Then, full papers were reviewed to identify all trials meeting the eligibility criteria. Disagreements were resolved through discussion or by consensus with a third investigator. When there were multiple articles from the same trial, the most completely or recently reported data were included.

Methodological quality and risk

Methodological quality of included articles was assessed using modified Jadad criteria with an eight-item scale (randomization, method of randomization, blinding, method of blinding, withdrawals and dropouts, inclusion and exclusion criteria, adverse effects and statistical analysis) by two independent reviewers.[25] Scores ranged from 0 to 8 (a high score indicating high quality). Scores of 4–8 denoted high quality and 0–3 poor or low quality.

Meta-analysis

The analysis of efficacy was based on the proportion of patients with a 75%, 90% and 100% improvement from baseline in the PASI score and the proportion of patients who achieved a DLQI score of 0 or 1. We evaluated safety by reviewing AEs, including any AEs, serious adverse events (SAEs), infection and cardiovascular disease (CVD). Additionally, discontinued treatment due to an AE or unsatisfactory therapeutic effect was analysed to assess safety and efficacy, respectively.

The statistical analysis was performed using Review Manager 5.2 (The Nordic Cochrane Centre, Copenhagen, Denmark) from the Cochrane Collaboration. All analysis indicators, which were categorical dichotomous variables, were assessed using relative risks (RR). $P < 0.05$ was considered to be statistically significant, and 95% confidence intervals (CI) were reported. Homogeneity was tested by the I^2 statistic (significance level at $I^2 > 50\%$).[26] A random-effects model was used if the I^2 statistic was significant. Otherwise, a fixed-effects model was used. To assess the potential confounding effect of heterogeneity, subgroup analyses for each of the IL-17 antagonists were performed.

Results

Literature search

A total of 405 relevant articles were identified from various databases, of which 359 articles were excluded after reviewing the titles and abstracts. Then, an additional 37 articles were excluded due to various reasons (not RCTs, not phase 3 trials or no required data). Finally, nine articles[27–35] including nine RCTs

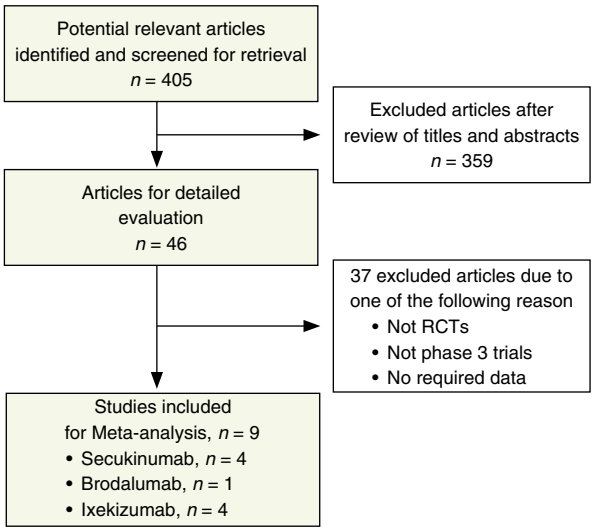

Figure 1 Flow chart demonstrating process of study selection.

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fully met the eligibility criteria (Fig. 1). Four articles[27–30] including five trials compared the effects of secukinumab with placebo, as the study by Langley *et al.* analysed two trials in one paper. One trial[31] evaluated the effects of brodalumab, and four articles[32–35] including three RCTs evaluated the effects of ixekizumab vs. placebo. The pooled analysis included 5951 patients with moderate-to-severe plaque psoriasis (4216 randomized to IL-17 antagonists and 1735 to placebo). The majority of studies retained were allocated high-quality scores (mean modified Jadad score 7.875 for included studies, all of the nine RCTs had a score ≥7). Included studies, basic characteristics of enrolled patients, details of drug therapy and modified Jadad scores are summarized in Table 1.

Efficacy

PASI 75 response Nine RCTs with 5941 patients reported the proportion of patients with a 75% improvement from baseline in the PASI score (PASI 75) at 12 weeks. IL-17 antagonists were associated with a higher PASI 75 response rate than the placebo group (RR = 18.85, 95% CI: 15.04–23.63, $P < 0.00001$) with no significant heterogeneity between studies ($P = 0.36$, $I^2 = 9\%$; Fig. 2).

PASI 90 response Nine RCTs with 5941 patients reported PASI 90 response rates. IL-17 antagonists were associated with a higher rate of PASI 90 responders than the placebo group (RR = 48.32, 95% CI: 31.48–74.15, $P < 0.00001$) with no significant heterogeneity between studies ($P = 0.39$, $I^2 = 6\%$; Fig. 3).

PASI 100 response Nine RCTs with 5941 patients reported the PASI 100 response rates at 12 weeks. IL-17 antagonists were associated with a higher rate of PASI 100 responders than the placebo group (RR = 72.06, 95% CI: 34.13–152.16, $P < 0.00001$) without significant heterogeneity between studies ($P = 0.64$, $I^2 = 0\%$; Fig. 4).

DLQI score of 0 or 1 response rate Five RCTs with 3625 patients reported the DLQI score of 0 or 1 response rate at 12 weeks. The IL-17 antagonists group was associated with a higher rate of DLQI 0 or 1 response than the placebo group (RR = 7.11, 95% CI: 4.18–12.09, $P < 0.00001$) with significant heterogeneity between studies ($P = 0.0005$, $I^2 = 80\%$; Fig. 5). No studies reporting DLQI response for brodalumab were found.

Safety

The meta-analysis results for safety showed that the IL-17 antagonists were associated with a higher proportion of patients with any AEs (RR = 1.22, 95% CI: 1.15–1.29, $P < 0.00001$) with no significant heterogeneity between studies ($P = 0.49$, $I^2 = 0\%$; Fig. 6), but no increase in the proportion of patients with SAEs (OR = 1.20, 95% CI: 0.75–1.92, $P = 0.46$), with no significant heterogeneity between studies ($P = 0.97$, $I^2 = 0\%$), compared with the placebo group (Fig. 7).

The interest in AEs that accompany IL-17 treatment for psoriasis focuses on infection and CVD. The most commonly reported infection events were nasopharyngitis, upper respiratory tract infection (URTI), viral URTI, pharyngitis, sinusitis or rhinitis. IL-17 antagonists were associated with a higher infection proportion than the placebo group (RR = 1.29, 95% CI: 1.06–1.57, $P = 0.01$) with significant heterogeneity ($P = 0.02$, $I^2 = 56\%$; Fig. 8). Only four studies with 2143 patients receiving IL-17 antagonists and 700 patients receiving placebo reported CVD. The results showed no significant difference between the IL-17 antagonists group and the placebo group (RR = 1.30, 95% CI: 0.35–4.91, $P = 0.70$). There was no significant heterogeneity between studies ($P = 0.15$, $I^2 = 47\%$; Fig. 9).

In addition, we need to pay attention on the brodalumab. There are serious safety concerns involving suicide with this drug. In our included articles,[31] four patients died from suicide after receiving brodalumab. However, the investigator reported that there was no reasonable possibility that the suicide was related to brodalumab.

Discontinuations

The most common reasons for discontinuations were unsatisfactory therapeutic effect or unable to tolerate AEs. In five RCTs, significantly fewer patients receiving IL-17 antagonists were withdrawn due to unsatisfactory efficacy than patients receiving placebo (RR = 0.20, 95% CI: 0.08–0.49, $P = 0.0004$). In eight RCTs, the rates of discontinuations due to AEs did not differ significantly between IL-17 antagonists and placebo (RR = 1.16, 95% CI: 0.65–2.07, $P = 0.62$; Fig. 10).

Discussion

The meta-analysis results of phase 3 clinical trials indicate that IL-17 antagonists are effective agents in the treatment of patients with psoriasis. At week 12, a significantly greater percentage of patients receiving IL-17 antagonists compared with placebo achieved PASI 75, PASI 90, PASI 100 and DLQI 0/1. Although IL-17 antagonists had a higher rate of any AEs and infection than placebo, there was no significant difference in SAEs and discontinuations due to AEs. This suggests that IL-17 antagonists are well tolerated.

Normally, achieving PASI 75 has been regarded as a successful treatment objective,[36] but now, achieving PASI 90 or PASI 100 has become a new therapeutic goal.[37] A study finding[38] showed that patients with moderate-to-severe plaque psoriasis who achieve PASI 90 or PASI 100 experience greater improvement in quality of life than PASI 75 responders. Our research showed IL-17 antagonists not only improved psoriasis symptoms and resulted in superior PASI response but also improved DLQI response rates compared with placebo. This finding further

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Table 1 Basic characteristics of included studies

Study	Journal	Dose and dosing schedule	No. of patients	Age (years)	Male patients *N* (%)	Duration of psoriasis (years)	PASI score	Study duration	Modified Jadad score
Secukinumab									
Paul (2015)[27]	JEADV		182					12 weeks	7
		150 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	61	43.9 ± 14.41	41(67.2)	20.6 ± 14.54	22.0 ± 8.85		
		300 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	60	46.6 ± 14.23	46(76.7)	21.0 ± 13.51	18.9 ± 6.37		
		Placebo at weeks 0, 1, 2, 3, 4, then every 4 weeks	61	43.7 ± 12.74	38(62.3)	19.86 ± 12.2	19.4 ± 6.70		
Ohtsuki (2014)[28]	JOD		87					12 weeks	8
		150 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	29	48.2 ± 13.08	23(79.3)	15.6 ± 10.41	28.2 ± 13.64		
		300 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	29	51.9 ± 11.77	26(89.7)	15.6 ± 10.30	26.7 ± 10.49		
		Placebo at weeks 0, 1, 2, 3, 4, then every 4 weeks	29	50.2 ± 13.62	23(79.3)	14.1 ± 10.91	21.4 ± 10.31		
Blauvelt (2015)[29]	BJD		177					12 weeks	8
		150 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	59	46.0 ± 15.09	40(67.8)	20.4 ± 12.97	20.5 ± 8.29		
		300 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	59	45.1 ± 12.57	38(64.4)	18.0 ± 11.86	20.7 ± 7.95		
		Placebo at weeks 0, 1, 2, 3, 4, then every 4 weeks	59	46.5 ± 14.14	39(66.1)	20.2 ± 14.22	21.1 ± 8.49		
Langley (2014) ERASURE[30]	NEJM		738					12 weeks	8
		150 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	245	44.9 ± 13.3	168(68.6)	17.5 ± 12.0	22.3 ± 9.8		
		300 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	245	44.9 ± 13.5	169(69.0)	17.4 ± 11.1	22.5 ± 9.2		
		Placebo at weeks 0, 1, 2, 3, 4, then every 4 weeks	248	45.4 ± 12.6	172(69.4)	17.3 ± 12.4	21.4 ± 9.1		
Langley (2014)FIXTURE[30]	NEJM		980					12 weeks	8
		150 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	327	45.4 ± 12.9	236(72.2)	17.3 ± 12.2	23.7 ± 10.5		
		300 mg sc at weeks 0, 1, 2, 3, 4, then every 4 weeks	327	44.5 ± 13.2	224(68.5)	15.8 ± 12.3	23.9 ± 9.9		
		Placebo at weeks 0, 1, 2, 3, 4, then every 4 weeks	326	44.1 ± 12.6	237(72.7)	16.6 ± 11.6	24.1 ± 10.5		

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Table 1 *Continued*

Study	Journal	Dose and dosing schedule	No. of patients	Age (years)	Male patients *N* (%)	Duration of psoriasis (years)	PASI score	Study duration	Modified Jadad score
Brodalumab									
Papp (2016)[31]	BJD		661					12 weeks	8
		140 mg every 2 weeks, with an additional dose at week 1	219	46 ± 13	162(74)	19 ± 13	20.0 ± 7.4		
		210 mg every 2 weeks, with an additional dose at week 1	222	46 ± 12	161(73)	20 ± 13	19.4 ± 6.6		
		Placebo	220	47 ± 13	161(73)	21 ± 12	19.7 ± 7.7		
Ixekizumab									
UNCOVER-1			1296					12 weeks	8
		80 mg every 2 weeks with a starting dose of 160 mg	433	45.1 ± 12.4	291(67.2)	19.9 ± 11.9	20.1 ± 8.0		
		80 mg every 4 weeks with a starting dose of 160 mg	432	45.6 ± 13.0	289(66.9)	19.5 ± 11.9	20.0 ± 7.3		
		Placebo	431	46.4 ± 13.4	303(70.3)	19.5 ± 11.7	20.3 ± 8.6		
UNCOVER-2			866					12 weeks	8
		80 mg every 2 weeks with a starting dose of 160 mg	351	44.5 ± 13.3	221(63.0)	18.3 ± 12.1	19.4 ± 7.3		
		80 mg every 4 weeks with a starting dose of 160 mg	347	45.0 ± 13.5	244(70.3)	18.5 ± 12.7	20.0 ± 7.0		
		Placebo	168	45.3 ± 12.1	120(71.4)	19.1 ± 12.7	20.6 ± 8.4		
UNCOVER-3			964					12 weeks	8
		80 mg every 2 weeks with a starting dose of 160 mg	385	45.6 ± 13.1	254(66.0)	17.8 ± 12.2	20.7 ± 8.2		
		80 mg every 4 weeks with a starting dose of 160 mg	386	45.6 ± 12.8	258(66.8)	18.5 ± 12.5	21.2 ± 8.1		
		Placebo	193	46.4 ± 12.1	137(71.0)	18.2 ± 12.5	21.1 ± 8.4		

STM, Science Translational Medicine; BJD, British Journal of Dermatology; JEADV, Journal of the European Academy of Dermatology and Venereology; JID, Journal of Investigative Dermatology; JOD, Journal of Dermatology; NEJM, New England Journal of Medicine; JACI, Journal of Allergy and Clinical Immunology; iv, intravenous; sc, subcutaneous; NR, not reported; PASI, Psoriasis Area and Severity Index.

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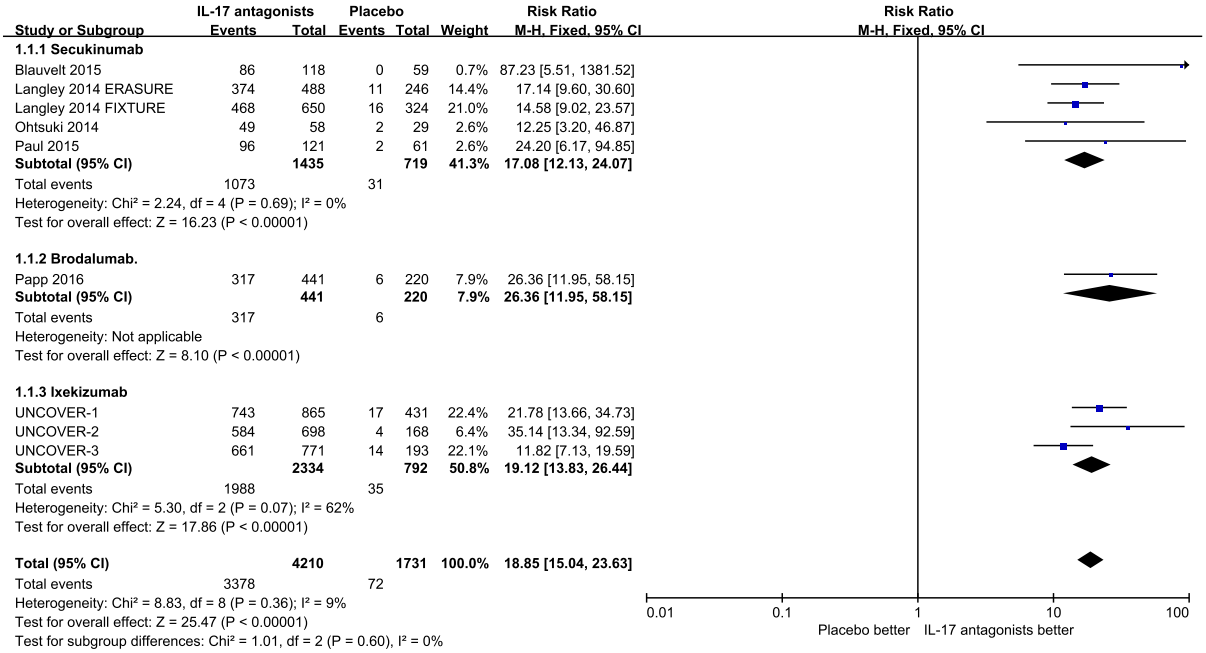

Figure 2 Meta-analysis of the rate of the PASI 75 responders.

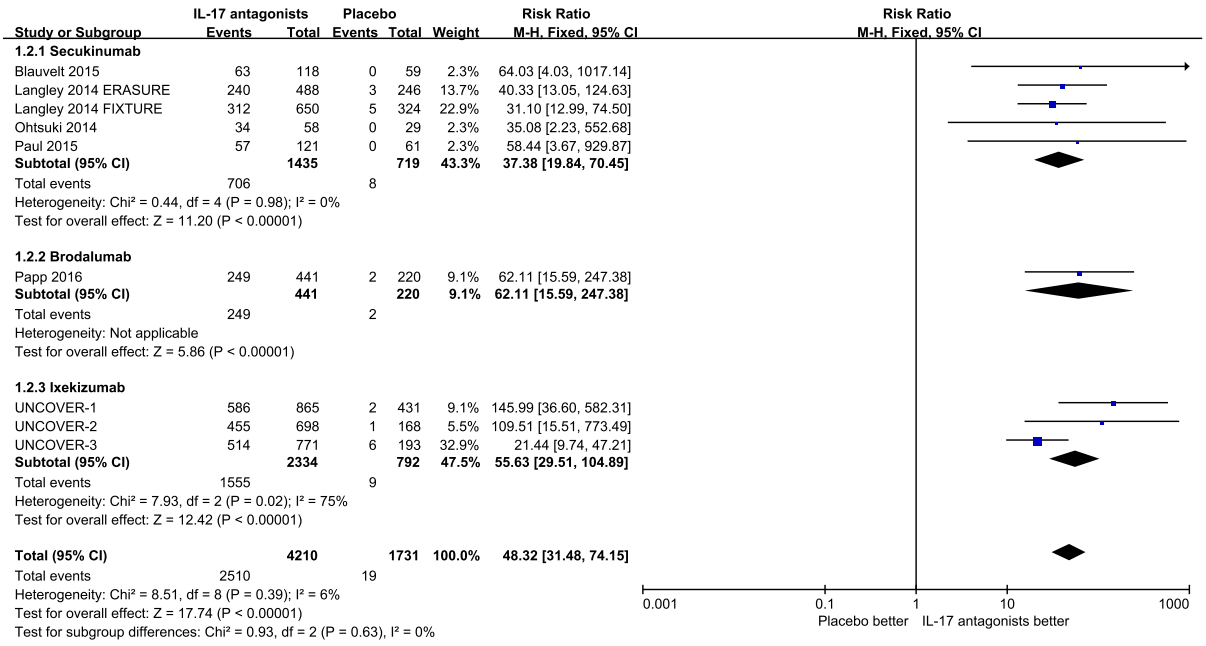

Figure 3 Meta-analysis of the rate of the PASI 90 responders.

certifies the importance of achieving nearly clear or clear skin by week 12 and confirms the effect of healthy skin on quality of life.

In addition, it should be noted there was a higher incidence of discontinuations due to lack of efficacy in placebo groups compared with IL-17 antagonist groups, which suggests that the IL-17 antagonists possibly show significant positive benefits. The findings are similar to other trials.[39,40]

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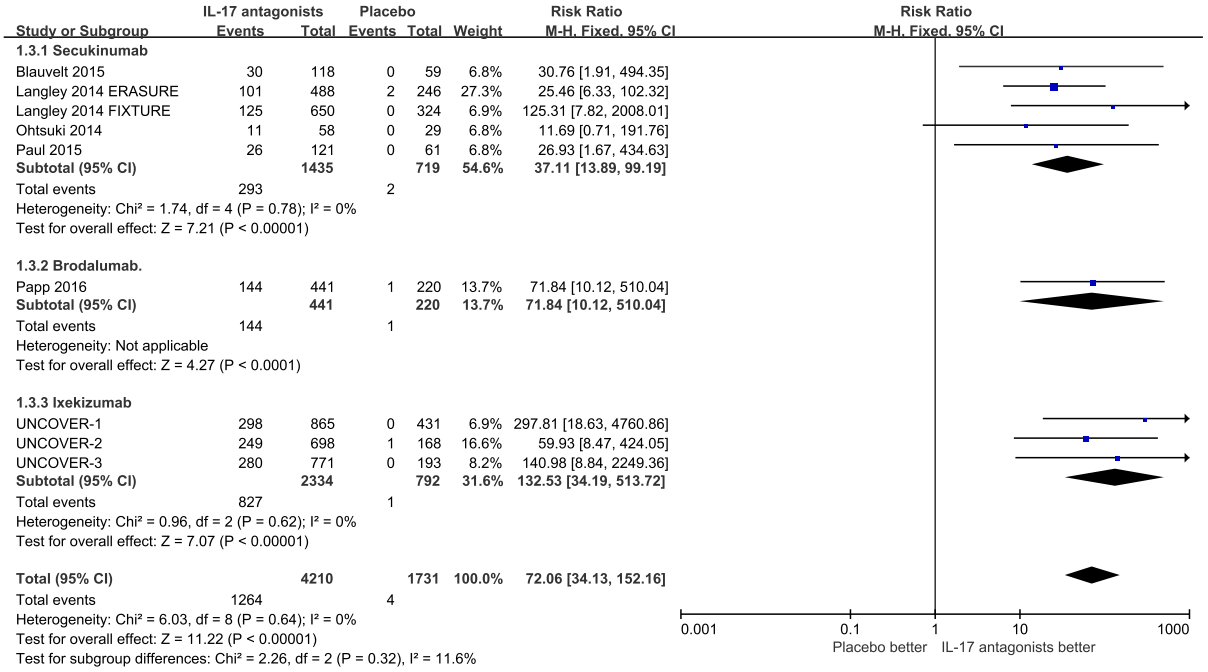

Figure 4 Meta-analysis of the rate of the PASI 100 responders.

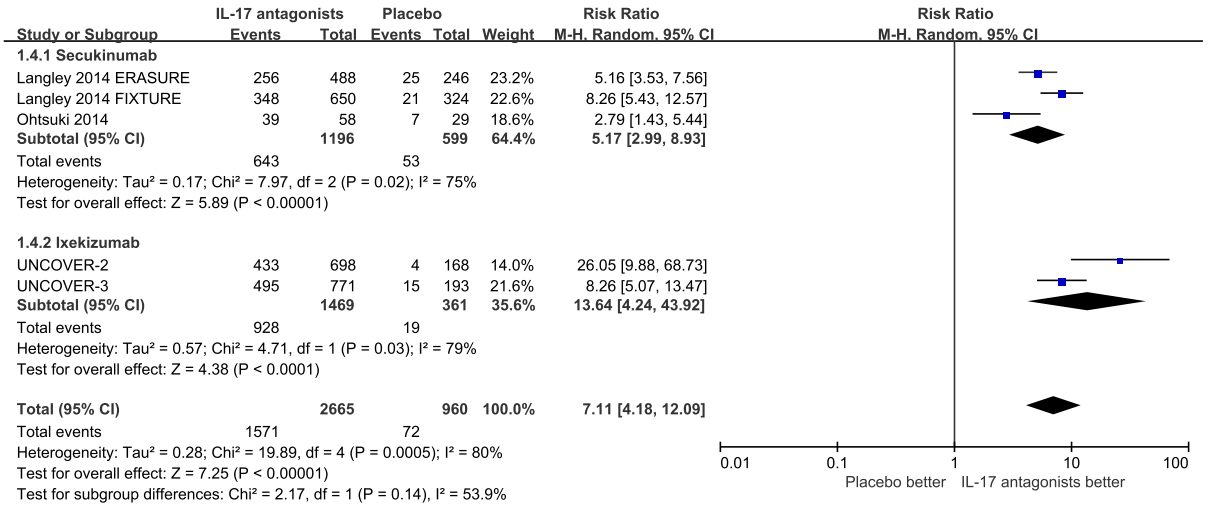

Figure 5 Meta-analysis of the proportion of patients who had a DLQI 0 or 1.

Safety analysis showed IL-17 antagonists had a higher rate of any AEs but there was no increase in the incidence of SAEs when compared with placebo. Most of the AEs were mild or moderate in severity. Compared with placebo, IL-17 antagonists were not associated with higher discontinuations due to AEs. Thus, these data indicate that IL-17 antagonists are well tolerated.

Infections mentioned in this study primarily included candida infection, nasopharyngitis, URTI, viral URTI, pharyngitis, rhinitis. IL-17A plays a key role in host defence, specifically in mucocutaneous microbial surveillance[41] and stimulation of granulopoiesis and neutrophil trafficking.[42] Vigilance with the potential for candida infection or neutropenia will be necessary for patients administered IL-17 antagonists. In our meta-

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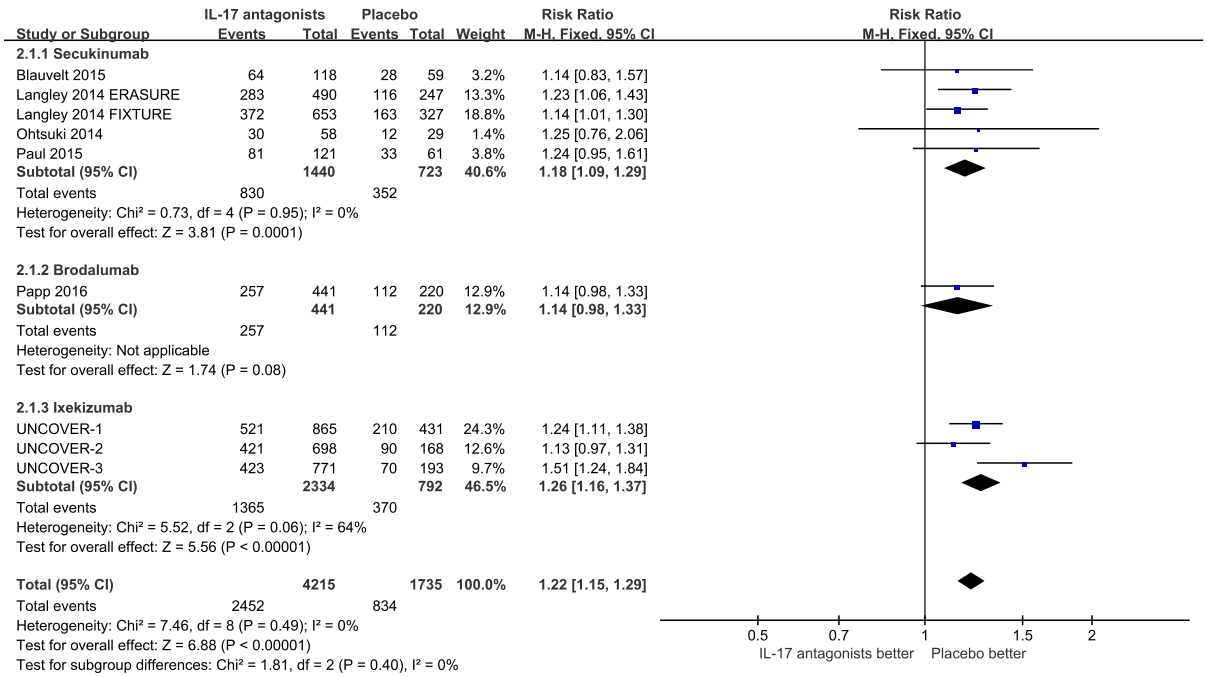

Figure 6 Meta-analysis of the proportion of patients with any adverse events.

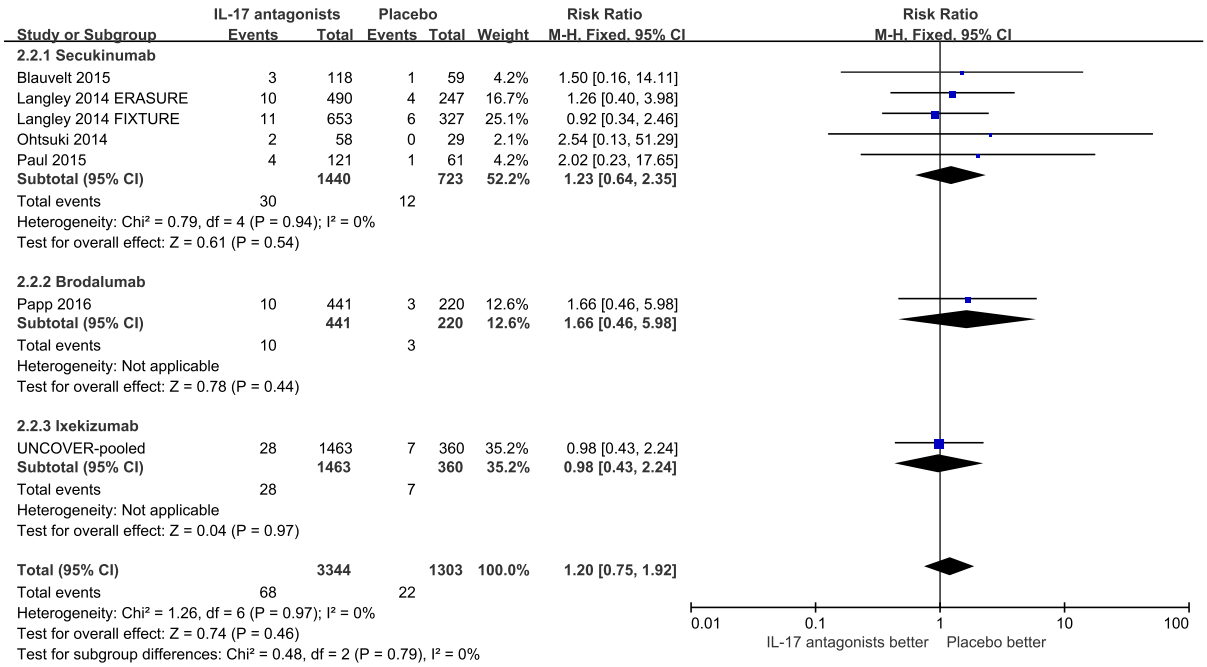

Figure 7 Meta-analysis of the proportion of patients with serious adverse events.

analysis, the IL-17 antagonist group was associated with a higher infection rate, but this rarely led to withdrawal from the study, and most cases were mild or moderate.

Plaque psoriasis is associated with a higher risk of CVD.[43,44] Epidemiologic evidence indicates that systemic therapy can decrease the risk of CVD by reducing inflammation.[45] Elevated

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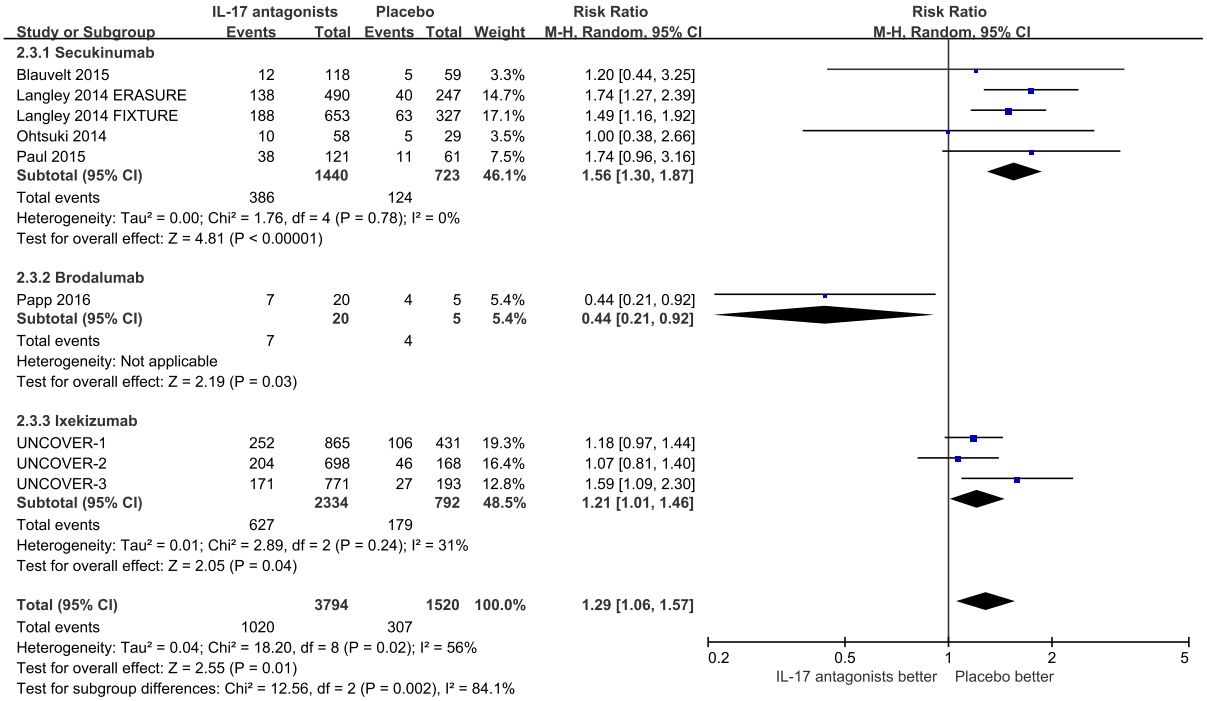

Figure 8 Meta-analysis of the risk of infection.

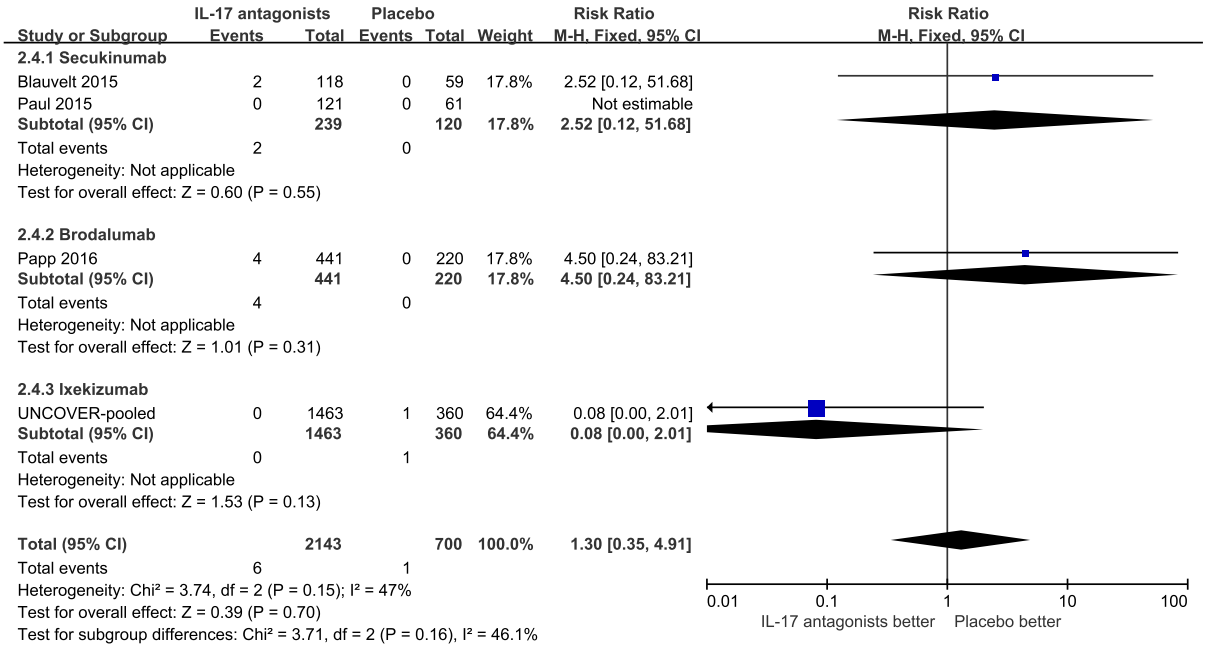

Figure 9 Meta-analysis of the risk of cardiovascular disease.

levels of IL-17A/IL-17F were measured in atherosclerotic lesions and advanced plaque.[46] Patients with unstable angina and acute myocardial infarction also had higher IL-17 levels, which drive inflammation to promote plaque progression and instability.[47] We may infer that IL-17 antagonists could possibly play an outstanding role in reducing cardiovascular morbidity. Findings

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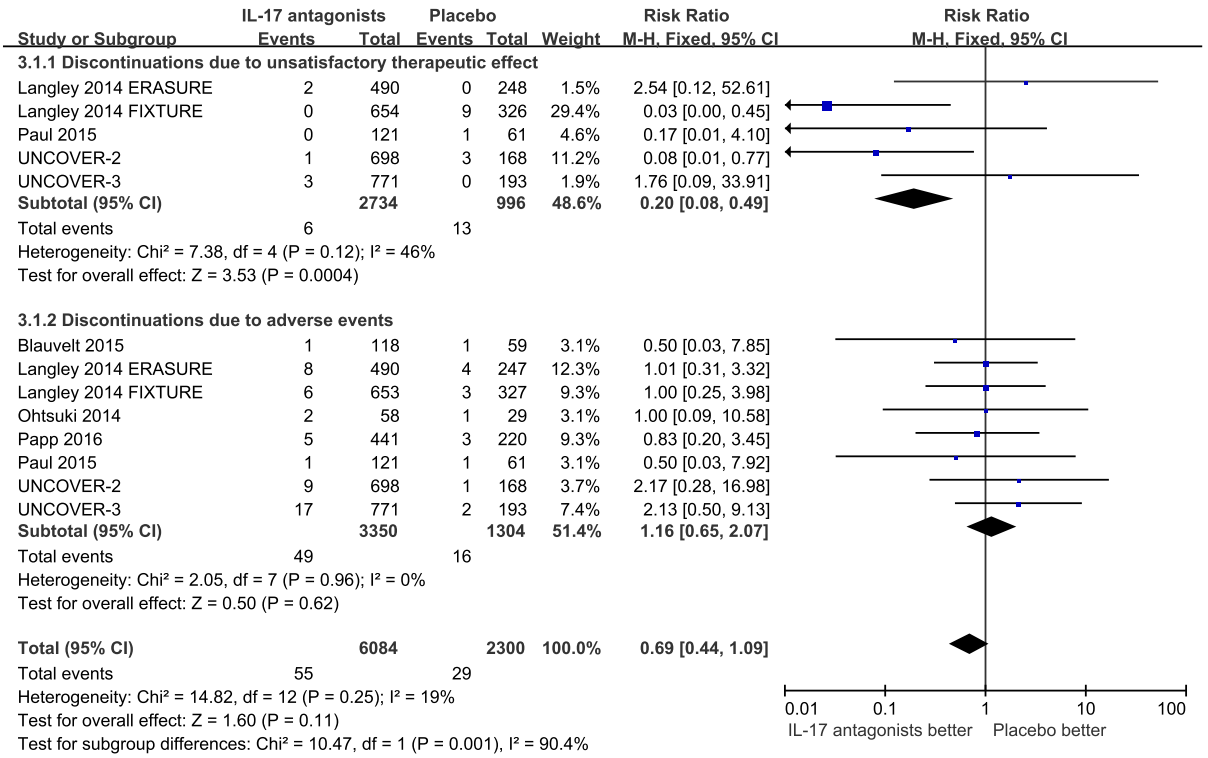

Figure 10 Meta-analysis of incidence of discontinuations.

from this meta-analysis indicate that IL-17 antagonists do not increase the incidence of CVD, and large sample clinical trials are needed to determine whether this targeted anti-IL-17 therapy will decrease the risk of CVD and provide a promising alternative for psoriasis patients with heart disease.

Both secukinumab and ixekizumab are approved by Food and Drug Administration (FDA), while brodalumab remains in development. In fact, four patients with psoriasis died from suicide after receiving brodalumab. Although concerning, the suicides have no a causative relationship with brodalumab, especially given that patients with psoriasis are already at a high risk of depression and suicide attempt. The long-term pharmacovigilance will help resolve this issue.

One limitation of our study was that we did not compare the efficacies or AEs between different IL-17 antagonist doses or different end points. However, from the included study results, we found that the secukinumab 300-mg dose was more effective than the 150-mg dose, suggesting that there is a dose–response effect. We also found the PASI score was significantly improved as early as 1–4 weeks in IL-17 antagonists[28,48–52] compared to placebo. These rapid effects often occurred in the higher-dose groups, which suggests that IL-17 antagonists may be more suitable for patients who want to improve the clinical symptoms and quality of life as soon as possible. Another limitation was that IL-17 antagonists are in ongoing clinical trials and not widely used in clinical treatment for patients with psoriasis. Hence, more high-quality, large prospective RCTs with long-term follow-up are needed.

In summary, IL-17 antagonists were effective and well tolerated. They could provide a credible alternative for patients with moderate-to-severe psoriasis, or even those with heart disease. Vigilance with the potential for infection will be necessary for patients administered IL-17 antagonists.

Reference

1 Jacobson CC, Kumar S, Kimball AB. Latitude and psoriasis prevalence. *J Am Acad Dermatol* 2011; **65**: 870–873.
2 de Korte J, Sprangers MA, Mombers FM, Bos JD. Quality of life in patients with psoriasis: a systematic literature review. *J Investig Dermatol Symp Proc* 2004; **9**: 140–147.
3 Stern RS, Nijsten T, Feldman SR, Margolis DJ, Rolstad T. Psoriasis is common, carries a substantial burden even when not extensive, and is associated with widespread treatment dissatisfaction. *J Investig Dermatol Symp Proc* 2004; **9**: 136–139.
4 Pathirana D, Ormerod AD, Saiag P *et al.* European S3-guidelines on the systemic treatment of psoriasis vulgaris. *J Eur Acad Dermatol Venereol* 2009; **23**(Suppl 2): 1–70.
5 Reich K, Nestle FO, Papp K *et al.* Infliximab induction and maintenance therapy for moderate-to-severe psoriasis: a phase III, multicentre, double-blind trial. *Lancet* 2005; **366**: 1367–1374.

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6 Papp KA, Poulin Y, Bissonnette R *et al.* Assessment of the long-term safety and effectiveness of etanercept for the treatment of psoriasis in an adult population. *J Am Acad Dermatol* 2012; **66**: e33–e45.
7 Menter A, Tyring SK, Gordon K *et al.* Adalimumab therapy for moderate to severe psoriasis: a randomized, controlled phase III trial. *J Am Acad Dermatol* 2008; **58**: 106–115.
8 Ellis CN, Krueger GG; Alefacept Clinical Study Group. Treatment of chronic plaque psoriasis by selective targeting of memory effector T lymphocytes. *N Engl J Med* 2001; **345**: 248–255.
9 Leonardi CL, Kimball AB, Papp KA *et al.* Efficacy and safety of ustekinumab, a human interleukin-12/23 monoclonal antibody, in patients with psoriasis: 76-week results from a randomised, double-blind, placebo-controlled trial (PHOENIX 1). *Lancet* 2008; **371**: 1665–1674.
10 Krueger GG, Langley RG, Leonardi C *et al.* A human interleukin-12/23 monoclonal antibody for the treatment of psoriasis. *N Engl J Med* 2007; **356**: 580–592.
11 Lowes MA, Bowcock AM, Krueger JG. Pathogenesis and therapy of psoriasis. *Nature* 2007; **445**: 866–873.
12 Miossec P, Korn T, Kuchroo VK. Interleukin-17 and type 17 helper T cells. *N Engl J Med* 2009; **361**: 888–898.
13 Lowes MA, Kikuchi T, Fuentes-Duculan J *et al.* Psoriasis vulgaris lesions contain discrete populations of Th1 and Th17 T cells. *J Invest Dermatol* 2008; **128**: 1207–1211.
14 Pène J, Chevalier S, Preisser L *et al.* Chronically inflamed human tissues are infiltrated by highly differentiated Th17 lymphocytes. *J Immunol* 2008; **180**: 7423–7430.
15 Arican O, Aral M, Sasmaz S, Ciragil P. Serum levels of TNF-alpha, IFN-gamma, IL-6, IL-8, IL-12, IL-17, and IL-18 in patients with active psoriasis and correlation with disease severity. *Mediators Inflamm* 2005; **2005**: 273–279.
16 Li J, Chen X, Liu Z, Yue Q, Liu H. Expression of Th17 cytokines in skin lesions of patients with psoriasis. *J Huazhong Univ Sci Technolog Med Sci* 2007; **27**: 330–332.
17 Bowcock AM, Krueger JG. Getting under the skin: the immunogenetics of psoriasis. *Nat Rev Immunol* 2005; **5**: 699–711.
18 Chiricozzi A, Guttman-Yassky E, Suárez-Fariñas M *et al.* Integrative responses to IL-17 and TNF-α in human keratinocytes account for key inflammatory pathogenic circuits in psoriasis. *J Invest Dermatol* 2011; **131**: 677–687.
19 Wilson NJ, Boniface K, Chan JR *et al.* Development, cytokine profile and function of human interleukin 17-producing helper T cells. *Nat Immunol* 2007; **8**: 950–957.
20 Cai Y, Shen X, Ding C *et al.* Pivotal role of dermal IL-17-producing γδ T cells in skin inflammation. *Immunity* 2011; **35**: 596–610.
21 van den Berg WB, McInnes IB. Th17 cells and IL-17 a–focus on immunopathogenesis and immunotherapeutics. *Semin Arthritis Rheum* 2013; **43**: 158–170.
22 Waisman A. To be 17 again–anti-interleukin-17 treatment for psoriasis. *N Engl J Med* 2012; **366**: 1251–1252.
23 Chiricozzi A, Krueger JG. IL-17 targeted therapies for psoriasis. *Expert Opin Investig Drugs* 2013; **22**: 993–1005.
24 Tausend W, Downing C, Tyring S. Systematic Review of Interleukin-12, Interleukin-17, and Interleukin-23 Pathway Inhibitors for the Treatment of Moderate-to-Severe Chronic Plaque Psoriasis: Ustekinumab, Briakinumab, Tildrakizumab, Guselkumab, Secukinumab, Ixekizumab, and Brodalumab. *J Cutan Med Surg* 2014; **18**: 1–14.
25 Oremus M, Wolfson C, Perrault A *et al.* Interrater reliability of the modified Jadad quality scale for systematic reviews of Alzheimer's disease drug trials. *Dement Geriatr Cogn Disord* 2001; **12**: 232–236.
26 Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002; **21**: 1539–1558.
27 Paul C, Lacour JP, Tedremets L *et al.* Efficacy, safety and usability of secukinumab administration by autoinjector/pen in psoriasis: a randomized, controlled trial (JUNCTURE). *J Eur Acad Dermatol Venereol* 2015; **29**: 1082–1090.
28 Ohtsuki M, Morita A, Abe M *et al.* Secukinumab efficacy and safety in Japanese patients with moderate-to-severe plaque psoriasis: subanalysis from ERASURE, a randomized, placebo-controlled, phase 3 study. *J Dermatol* 2014; **41**: 1039–1046.
29 Blauvelt A, Prinz JC, Gottlieb AB *et al.* Secukinumab administration by pre-filled syringe: efficacy, safety and usability results from a randomized controlled trial in psoriasis (FEATURE). *Br J Dermatol* 2015; **172**: 484–493.
30 Langley RG, Elewski BE, Lebwohl M *et al.* Secukinumab in plaque psoriasis–results of two phase 3 trials. *N Engl J Med* 2014; **371**: 326–338.
31 Papp KA, Reich K, Paul C *et al.* A prospective phase III, randomized, double-blind, placebo-controlled study of brodalumab in patients with moderate-to-severe plaque psoriasis. *Br J Dermatol* 2016; **175**: 273–286.
32 Armstrong AW, Lynde CW, McBride SR *et al.* Effect of ixekizumab treatment on work productivity for patients with moderate-to-severe plaque psoriasis: analysis of results from 3 randomized phase 3 clinical trials. *JAMA Dermatol* 2016; **152**: 661–669.
33 Griffiths CE, Reich K, Lebwohl M *et al.* Comparison of ixekizumab with etanercept or placebo in moderate-to-severe psoriasis (UNCOVER-2 and UNCOVER-3): results from two phase 3 randomised trials. *Lancet* 2015; **386**: 541–551.
34 Farahnik B, Beroukhim K, Zhu TH *et al.* Ixekizumab for the treatment of psoriasis: a review of phase III trials. *Dermatol Ther (Heidelb)* 2016; **6**: 25–37.
35 Gordon KB, Blauvelt A, Papp KA *et al.* Phase 3 trials of ixekizumab in moderate-to-severe plaque psoriasis. *N Engl J Med* 2016; **375**: 345–356.
36 Menter A, Korman NJ, Elmets CA *et al.* Guidelines of care for the management of psoriasis and psoriatic arthritis: section 6. Guidelines of care for the treatment of psoriasis and psoriatic arthritis: case-based presentations and evidence-based conclusions. *J Am Acad Dermatol* 2011; **65**: 137–174.
37 Puig L. PASI90 response: the new standard in therapeutic efficacy for psoriasis. *J Eur Acad Dermatol Venereol* 2015; **29**: 645–648.
38 Revicki DA, Willian MK, Menter A, Saurat JH, Harnam N, Kaul M. Relationship between clinical response to therapy and health-related quality of life outcomes in patients with moderate to severe plaque psoriasis. *Dermatology* 2008; **216**: 260–270.
39 Paul C, Reich K, Gottlieb AB *et al.* Secukinumab improves hand, foot and nail lesions in moderate-to-severe plaque psoriasis: subanalysis of a randomized, double-blind, placebo-controlled, regimen-finding phase 2 trial. *J Eur Acad Dermatol Venereol* 2014; **28**: 1670–1675.
40 Gordon KB, Leonardi CL, Lebwohl M *et al.* A 52-week, open-label study of the efficacy and safety of ixekizumab, an anti-interleukin-17A monoclonal antibody, in patients with chronic plaque psoriasis. *J Am Acad Dermatol* 2014; **71**: 1176–1182.
41 Puel A, Cypowyj S, Bustamante J *et al.* Chronic mucocutaneous candidiasis in hu -mans with inborn errors of interleukin-17 immunity. *Science* 2011; **332**: 65–68.
42 Krstic A, Mojsilovic S, Jovcic G, Bugarski D. The potential of interleukin-17 to mediate hematopoietic response. *Immunol Res* 2012; **52**: 34–41.
43 Samarasekera EJ, Neilson JM, Warren RB, Parnham J, Smith CH. Incidence of cardiovascular disease in individuals with psoriasis: a systematic review and meta-analysis. *J Invest Dermatol* 2013; **133**: 2340–2346.
44 Miller IM, Skaaby T, Ellervik C, Jemec GB. Quantifying cardiovascular disease risk factors in patients with psoriasis: a meta-analysis. *Br J Dermatol* 2013; **169**: 1180–1187.
45 Wu JJ, Poon KY, Bebchuk JD. Association between the type and length of tumor necrosis factor inhibitor therapy and myocardial infarction risk in patients with psoriasis. *J Drugs Dermatol* 2013; **12**: 899–903.
46 de Boer OJ, van der Meer JJ, Teeling P *et al.* Differential expression of interleukin-17 family cytokines in intact and complicated human atherosclerotic plaques. *J Pathol* 2010; **220**: 499–508.
47 Hashmi S, Zeng QT. Role of interleukin-17 and interleukin-17-induced cytokines interleukin-6 and interleukin-8 in unstable coronary artery disease. *Coron Artery Dis* 2006; **17**: 699–706.

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48 Hueber W, Patel DD, Dryja T *et al.* Effects of AIN457, a fully human antibody to interleukin-17A, on psoriasis, rheumatoid arthritis, and uveitis. *Sci Transl Med* 2010; **2**: 52ra72.
49 Rich P, Sigurgeirsson B, Thaci D *et al.* Secukinumab induction and maintenance therapy in moderate-to-severe plaque psoriasis: a randomized, double-blind, placebo-controlled, phase II regimen-finding study. *Br J Dermatol* 2013; **168**: 402–411.
50 Papp KA, Reid C, Foley P *et al.* Anti-IL-17 receptor antibody AMG 827 leads to rapid clinical response in subjects with moderate to severe psoriasis: results from a phase I, randomized, placebo-controlled trial. *J Invest Dermatol* 2012; **132**: 2466–2469.
51 Papp KA, Leonardi C, Menter A *et al.* Brodalumab, an anti-interleukin-17-receptor antibody for psoriasis. *N Engl J Med* 2012; **366**: 1181–1189.
52 Leonardi C, Matheson R, Zachariae C *et al.* Anti-interleukin-17 monoclonal antibody ixekizumab in chronic plaque psoriasis. *N Engl J Med* 2012; **366**: 1190–1199.

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